

Toric Varieties Modulo Reflections

arXiv:2410.14653

Tao Gong

Western University

Joint with Colin Crowley & Conner Simpson

CMS Summer Meeting

June 6, 2026

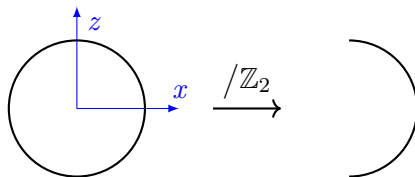
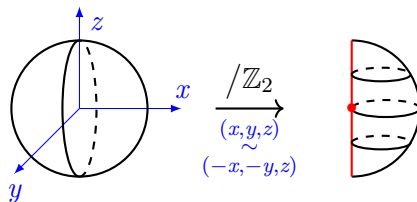
Table of Contents

1 Background

2 Complex Quotients

3 Real Quotients

Pseudo-Motivations



Generalization: S^2 is a complex toric variety, S^1 is a real toric variety, $\mathbb{Z}_2$ is a reflection group.

Motivations

Why toric topology?

- Algebraic geometry: Torus embedding [Demazure'70].
- Combinatorial geometry: Polytope $P \rightsquigarrow$ toric variety X_P [Stanley'80].
- Symplectic geometry: Atiyah–Guillemin–Sternberg convexity theorem.

Why reflection group?

- Weyl group W naturally shows up in Lie groups.
- Coxeter group in geometric group theory.

What is X_P with reflections?

- Generalized Losev–Manin moduli spaces.
- W -representation $H^*(X_P)$.
- K -theory of matroids.
- ...

Cones & Affine Toric Varieties

The **lattice** M has rank n ($M \cong \mathbb{Z}^n$), has dual lattice $N (= \text{hom}(M, \mathbb{Z}))$.

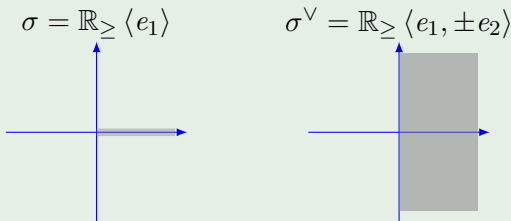
A point $u = (u_1, \dots, u_n) \in M$ defines a monomial $t^u = t_1^{u_1} t_2^{u_2} \cdots t_n^{u_n}$.

A **cone** $\sigma \subset N_{\mathbb{R}}$ is a non-negative span of vectors lying in N .

The **dual cone** $\sigma^{\vee} := \{x \in M_{\mathbb{R}} : \langle x, y \rangle \geq 0, \forall y \in \sigma\}$.

The **affine toric variety** $X_{\sigma} = \text{Spec}(\mathbb{C}[t^u : u \in \sigma^{\vee} \cap M])$.

Example



$$X_{\sigma} = \text{Spec}(\mathbb{C}[t_1, t_2^{\pm 1}]) \cong \mathbb{C} \times \mathbb{C}^*$$

Fans & Toric Varieties

A **fan** Σ is a set of cones, closed under face inclusions and intersections.
The **toric variety** $X_\Sigma = \bigcup_{\sigma \in \Sigma} X_\sigma$.

Example

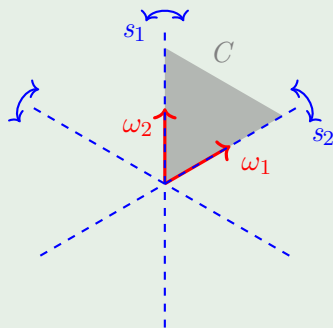
In $\mathbb{R}$, $P = [-1, 1]$ with normal fan Σ_P .

$$\begin{array}{ccccccc} \Sigma_P & & \xleftarrow{(-\infty, 0]} & \bullet & \xrightarrow{[0, \infty)} & & \\ X_\sigma & & \mathbb{C} & & \mathbb{C}^* & & \mathbb{C} \\ X_\sigma^{\mathbb{R}} & & \mathbb{R} & & \mathbb{R}^* & & \mathbb{R} \\ X_P = \operatorname{colim} & & \left(\mathbb{C} \xleftarrow{x \mapsto x^{-1}} \mathbb{C}^* \xrightarrow{x \mapsto x} \mathbb{C} \right) & & & & = \mathbb{CP}^1 \\ X_P^{\mathbb{R}} = \operatorname{colim} & & \left(\mathbb{R} \xleftarrow{x \mapsto x^{-1}} \mathbb{R}^* \xrightarrow{x \mapsto x} \mathbb{R} \right) & & & & = \mathbb{RP}^1 \end{array}$$

Weyl Groups

A **Weyl group** W is finite, generated by reflections, preserving a lattice.

Example (of type A_2)



$$W = \langle s_1, s_2 \rangle = S_3 \leq O(\mathbb{R}^2).$$

$$W_{\{1\}} = \langle s_1 \rangle.$$

$$C := \mathbb{R}_{\geq} \langle \omega_1, \omega_2 \rangle \cong \mathbb{R}^2 / W.$$

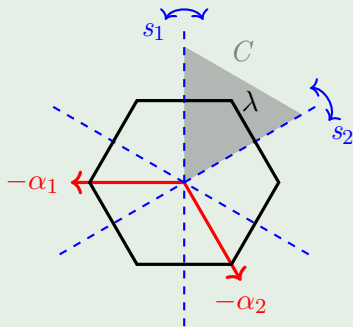
$$W \curvearrowright N := \mathbb{Z} \langle \omega_1, \omega_2 \rangle.$$

W -Permutohedra & Quotient Polytopes

The W -**permutohedron** $M_{\mathbb{R}} \supset P := \text{Conv}(W(\lambda))$ with $\lambda \in M \cap \text{Int}(C)$.

The **quotient polytope** P/W is $M_{\mathbb{R}} \supset P \cap C \cong_{\text{comb.}} [0, 1]^n$.

Example (of type A_2)



P is a hexagon.

P/W is a square.

$\{s(C) \mid s \in W\}$ generates Σ_P .

$\{C, -\alpha_1, -\alpha_2\}$ generates $\Sigma_{P/W}$.

What is the relation between X_P and $X_{P/W}$?

Table of Contents

1 Background

2 Complex Quotients

3 Real Quotients

Rational Cohomology Rings

- $P, P/W \subset M_{\mathbb{R}}$ have each vertex lie in exactly n facets.
- $\ell_F \in N$ is the primitive outward-pointing normal vector of the facet F .

Theorem (Danilov'78, Jurkiewicz'80)

The rational cohomology ring of X_P is

$$H^*(X_P) = \frac{\mathbb{Q}[T_F : F \text{ is a facet of } P]}{I + J},$$

where each T_F is of degree 2, and

$$I = \left\langle T_{F_{i_1}} T_{F_{i_2}} \cdots T_{F_{i_k}} \mid F_{i_1} \cap F_{i_2} \cap \cdots \cap F_{i_k} = \emptyset \right\rangle,$$

$$J = \left\langle \sum_{\text{all facets } F} \langle u, \ell_F \rangle T_F \mid u \in M \right\rangle.$$

Similar is $H^(X_{P/W})$.*

Cohomology Isomorphisms

Theorem (Horiguchi–Masuda–Shareshian–Song'24)

There is a ring isomorphism $\Phi : H^(X_{P/W}) \rightarrow H^*(X_P)^W \cong H^*(X_P/W)$.*

Example ($P = [-1, 1] \subset \mathbb{R}$)

$W = O(\mathbb{R}^1) = \langle x \mapsto -x \rangle$. Then $P/W = [0, 1]$, and $X_P \cong \mathbb{CP}^1 \cong X_{P/W}$.

$$\Phi = \text{id} : H^*(\mathbb{CP}^1) \rightarrow H^*(\mathbb{CP}^1) = H^*(\mathbb{CP}^1)^W.$$

[HMSS'24] proved Φ for Weyl group W of classical types and any parabolic subgroup W_K .

[Gui–Hu–Liu'25] proved Φ for Weyl group W of any type and $W_K = W$.

[Song'22] proved Φ for any W -symmetric polygon and any W_K .

Question (cf. [Horiguchi–Masuda–Shareshian–Song'24])

Is there an equivalence between X_{P/W_K} and X_P/W_K ?

[Blume'15] proved variety isomorphisms for types A, B, C with $W_K = W$.

[G'24] proved homotopy equivalences for all types.

In *Toric Varieties* by [Cox-Little-Schenck'11], define

$$C(P) := \text{Cone}(P \times \{1\}) \subset M_{\mathbb{R}} \times \mathbb{R}.$$

Theorem (CLS'11, Theorem 7.1.13)

If P is a lattice polytope, then $X_P \cong \text{Proj}(\mathbb{C}[C(P) \cap M \times \mathbb{Z}])$.
If P/W is a lattice polytope, then $X_{P/W} \cong \text{Proj}(\mathbb{C}[C(P/W) \cap M \times \mathbb{Z}])$.

By *Geometric Invariant Theory*,

$$X_{P/W} \cong \text{Proj}(\mathbb{C}[C(P) \cap M \times \mathbb{Z}]^W).$$

Theorem (Crowley–G.–Simpson'24)

There is an algebra isomorphism

$$\Psi : \mathbb{Z}[C(P/W) \cap M \times \mathbb{Z}] \rightarrow \mathbb{Z}[C(P) \cap M \times \mathbb{Z}]^W,$$
$$t^u \mapsto \sum_{v \in W(u)} t^v,$$

which is still true when W is replaced by W_K , or P replaced by a W -polytope.

Table of Contents

1 Background

2 Complex Quotients

3 Real Quotients

A Question

Question

What is $X_P^{\mathbb{R}}/W_K$ the quotient of the real points $X_P^{\mathbb{R}}$?

By [Henderson'12, Cho-Choi-Kaji'19], $H^*(X_P^{\mathbb{R}}/W; \mathbb{Q}) = \mathbb{Q}$ for types A, B .

Topological Models via $\sim$

Define $\mathbb{Z}_2^n := N \otimes_{\mathbb{Z}} \mathbb{Z}_2$. By [Jurkiewicz'81], there is a homeomorphism

$$X_P^{\mathbb{R}} \cong (\mathbb{Z}_2^n \times P) / \sim,$$

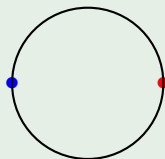
where $(t_1, p_1) \sim (t_2, p_2)$ iff $p_1 = p_2$, and $[t_1] = [t_2] \in \frac{N}{N \cap \langle \sigma_{p_1} \rangle} \otimes \mathbb{Z}_2$ with $\langle \sigma_{p_1} \rangle$ the $\mathbb{R}$ -span of the normal cone σ_{p_1} at p_1 .

Example ($P = [-1, 1] \subset \mathbb{R}$)

$$\mathbb{Z}_2 \times [-1, 1]$$



$$X_P^{\mathbb{R}} \cong S^1$$



Topological Models via $\approx$

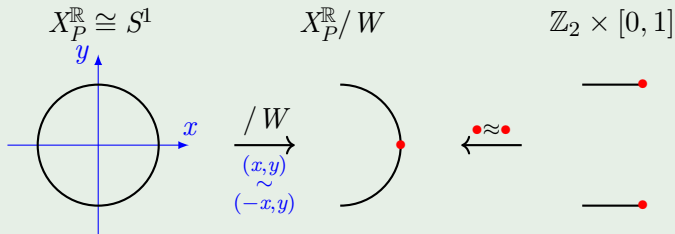
There is a homeomorphism

$$X_P^{\mathbb{R}} / W \cong \frac{\mathbb{Z}_2^n \times [0, 1]^n}{\approx} = \bigcup_{u \in \mathbb{Z}_2^n} \{u\} \times [0, 1]^n,$$

where $(t_1, p_1) \approx (t_2, p_2)$ iff $p_1 = p_2$ lying in the minimal face $H_I F_J := \{0\}^I \times \{1\}^J \times [0, 1]^c$ of $[0, 1]^n$, and $[t_1] = [t_2] \in \frac{\mathbb{Z}_2^n / \mathbb{Z}_2^J}{W_I}$.

Example (of type A_1)

In $\mathbb{R}$, $W = \langle s : x \mapsto -x \rangle$, $P = [-1, 1]$, and $P/W \cong [0, 1]$.



Filtrations

Order $\mathbb{Z}_2^n$ colexicographically in terms of $\{\omega_1, \omega_2, \dots, \omega_n\}$.

Write η_k for the vector of order k .

In particular, $\eta_0 = 0$, $\eta_{2^i-1} = \omega_i$, $\eta_{2^n-1} = \sum_{i=1}^n \omega_i$.

Filtrations

Order $\mathbb{Z}_2^n$ colexicographically in terms of $\{\omega_1, \omega_2, \dots, \omega_n\}$.

Write η_k for the vector of order k .

In particular, $\eta_0 = 0$, $\eta_{2^i-1} = \omega_i$, $\eta_{2^n-1} = \sum_{i=1}^n \omega_i$.

There are pushout diagrams, for all $k = 2^n - 2, 2^n - 3, \dots, 1, 0$,

$$\begin{array}{ccc} \mathbf{G}_{\eta_k} & \longrightarrow & \bigcup_{j \geq k+1} \{\eta_j\} \times [0, 1]^n \\ f_k \downarrow & & \lrcorner \quad \downarrow g_k \\ \{\eta_k\} \times [0, 1]^n & \longrightarrow & \bigcup_{j \geq k} \{\eta_j\} \times [0, 1]^n, \end{array}$$

Filtrations

Order $\mathbb{Z}_2^n$ colexicographically in terms of $\{\omega_1, \omega_2, \dots, \omega_n\}$.

Write η_k for the vector of order k .

In particular, $\eta_0 = 0$, $\eta_{2^i-1} = \omega_i$, $\eta_{2^n-1} = \sum_{i=1}^n \omega_i$.

There are pushout diagrams, for all $k = 2^n - 2, 2^n - 3, \dots, 1, 0$,

$$\begin{array}{ccc}
 \mathbf{G}_{\eta_k} & \longrightarrow & \bigcup_{j \geq k+1} \{\eta_j\} \times [0, 1]^n \\
 \downarrow f_k & & \downarrow g_k \\
 \{\eta_k\} \times [0, 1]^n & \longrightarrow & \bigcup_{j \geq k} \{\eta_j\} \times [0, 1]^n,
 \end{array}$$

If each f_k is a homotopy equivalence, then so is the pushout g_k .

$$\{\eta_{2^n-1}\} \times [0, 1]^n \xrightarrow{g_{2^n-2}} \bigcup_{j \geq 2^n-2} \{\eta_j\} \times [0, 1]^n \xrightarrow{g_{2^n-3}} \dots \xrightarrow{g_0} X_P^{\mathbb{R}} / W.$$

Unions of Faces

Each $\mathbf{G}_\eta$ is an **irreducible** union of proper faces of the cube $[0, 1]^n$

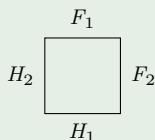
$$\mathbf{G}_\eta = H_{I_0} F_{J_0} \bigcup H_{I_1} F_{J_1} \bigcup \dots \bigcup H_{I_r} F_{J_r},$$

that is, each component is maximal and not contained in any other.

Lemma

If the component $H_{I_0} F_{J_0}$ intersects any other, then $\mathbf{G}_\eta \simeq pt.$

Example (of dimension 2)



$F_1 \cup F_2$



$H_1 \cup H_2 \cup F_1 \cup F_2$



$H_1 \cup F_1 \cup F_2$



$H_{12} \cup F_1 \cup F_2$

Theorem (Crowley–G.–Simpson'24)

Each $\mathbf{G}_\eta$ has a component that intersects any other. Then $X_P^{\mathbb{R}}/W \simeq pt$.

Proof Sketch.

An expression s of an element in W is called **effective on** $\eta \in \mathbb{Z}_2^n$ iff $s(\eta) \neq s'(\eta)$ for any sub-expression s' .

$H_I F_J$ is component $\Rightarrow s(\eta) + \sum_{j \in J} \omega_j^\vee > \eta$ for some effective $s \in W_I$.

Choose k not in $\text{Supp}(\eta) \cup \text{Supp}(s)$. Then $F_k \subset \mathbf{G}_\eta$ is the component meeting others.



Thank You!